
Is There a Universal ‘Sounds Like AI’ Direction? Probing and Steering AI Style in Transformer Residual Streams

Joanna Zhang
joannazhang@uchicago.edu

Abstract

1 Detection of AI-generated text has become increasingly important as large language
2 models are deployed at scale. Prior work has shown that high-level concepts such as
3 truthfulness, refusal, and sentiment are encoded as linear directions in the residual
4 streams of transformer models. We investigate whether “sounding like AI” — the
5 composite stylistic quality that makes LLM outputs identifiable — is similarly
6 represented, and whether steering along this direction can evade real-world AI
7 detectors. Using contrastive activation addition (CAA) on Qwen2.5-3B, we extract
8 an “AI style direction” from the HC3 dataset. Replicating prior work, we confirm
9 that this direction achieves 97.5% classification accuracy on ChatGPT-sourced
10 text, though 93% of the signal overlaps with text length. After controlling for
11 length, an 85.5% accurate residual style direction remains. Steering along this
12 direction reduces Pangram’s AI probability from 0.70 to 0.02 on ChatGPT-sourced
13 text, demonstrating near-complete evasion of a state-of-the-art detector. We further
14 extend the dataset to include Claude and Gemini responses, finding that these
15 models produce substantially longer outputs (97.5% and 96.1% length overlap
16 respectively), which largely dominates the extracted direction. To address this,
17 we propose a sentence-boundary truncation method that reduces length-direction
18 cosine similarity from 0.944 to 0.002, revealing a genuine cross-source style signal
19 at 95.8% accuracy. However, steering on this length-controlled combined dataset
20 produces inconsistent evasion results, suggesting that a universal model-agnostic
21 AI style direction may not be straightforwardly steerable. Our results highlight
22 both the promise and the limitations of residual stream manipulation as an approach
23 to AI detection evasion.

1 Introduction

24 The proliferation of large language models has intensified an ongoing arms race between AI text
25 generation and AI text detection. Tools such as GPTZero, ZeroGPT, and Pangram attempt to identify
26 machine-generated text by exploiting statistical and stylistic signatures in LLM outputs, while a
27 parallel line of work explores evasion strategies. Existing evasion approaches largely operate at
28 the surface level: recursive paraphrasing Sadasivan et al. [2023], post-hoc humanization, or prompt
29 engineering Mitchell et al. [2023]. These methods treat the model as a black box, modifying outputs
30 after the fact without any understanding of *why* the text sounds AI-generated in the first place. We
31 ask a more fundamental question: is “sounding like AI” encoded as an internal representation within
32 the model itself, and if so, can we manipulate that representation directly to evade detection?

34 A growing body of work in mechanistic interpretability has established that transformer language
35 models linearly encode high-level concepts as directions in their residual streams Marks and Tegmark
36 [2024], Arditì et al. [2024], Konen et al. [2024], and that these directions can be extracted and causally
37 steered via Contrastive Activation Addition Rinsky et al. [2024]. Whether “sounding like AI” — the

38 composite stylistic quality that makes LLM outputs identifiable — constitutes such a direction has
 39 not been directly investigated. Concurrently, Kuznetsov et al. (2025) use Sparse Autoencoders on
 40 residual streams to identify features contributing to AI text detection, finding that style and verbosity
 41 features are the most discriminative. Our work is complementary but distinct: we use the simpler
 42 difference-in-means approach and target *evasion* rather than detection interpretability.

43 We extract an “AI style direction” from Qwen2.5-3B using the HC3 dataset of paired human and
 44 ChatGPT responses, replicating prior work showing 97.5% classification accuracy with 93% overlap
 45 with text length. After controlling for length, a genuine residual style signal persists at 85.5%
 46 accuracy. Critically, steering along this direction reduces Pangram’s AI probability from 0.70 to 0.02,
 47 demonstrating near-complete evasion of a state-of-the-art detector through internal representation
 48 manipulation. We then extend to Claude and Gemini, finding that these models’ substantially greater
 49 verbosity dominates the extracted direction even more severely. To address this, we introduce a
 50 sentence-boundary truncation method that eliminates the length confound by construction, revealing
 51 a genuine cross-source style signal at 95.8% accuracy — but finding that steering on this combined
 52 dataset produces inconsistent evasion results, suggesting that a universal model-agnostic AI style
 53 direction may not be straightforwardly steerable.

54 Our main contributions are:

- 55 • We demonstrate near-complete evasion of Pangram (0.70 $\rightarrow$ 0.02 AI probability) via activa-
 56 tion steering on ChatGPT-sourced text, establishing residual stream manipulation as a viable
 57 attack vector against AI detectors.
- 58 • We extend the analysis to three AI sources (ChatGPT, Claude, Gemini), finding that verbosity
 59 accounts for 93–97.5% of the extracted direction across all sources and is substantially
 60 more severe for Claude and Gemini, suggesting that current AI style directions are largely
 61 verbosity detectors.
- 62 • After controlling for length via sentence-boundary truncation, we find a genuine cross-
 63 source style signal at 95.8% accuracy, but show that this direction does not steer cleanly
 64 across sources — suggesting AI style may be partially model-specific rather than a universal
 65 property of LLM-generated text.
- 66 • We find that CAA steering effectiveness decreases with model size across the Qwen2.5-
 67 Instruct family, with the 3B model reaching 0.219 Pangram probability vs. 0.741 and 0.605
 68 for 7B and 14B respectively.

69 2 Background

70 We build directly on a preliminary study Hypogenic-AI [2024] that applied CAA to AI text detection,
 71 which we summarize here as it serves as our baseline.

72 **Dataset.** The study used HC3 Guo et al. [2023], a corpus of paired human and ChatGPT responses to
 73 the same questions across five domains. After filtering to 50–1500 characters, 200 pairs were used
 74 for direction extraction, 50 for validation, and 100 for test. ChatGPT responses are systematically
 75 longer than human responses (mean 914 vs. 446 characters), a confound that substantially affects the
 76 analysis.

77 **Method.** For each layer ℓ of Qwen 2.5-3B, a difference-in-means direction is computed between AI
 78 and human last-token residual stream activations. This direction serves both as a linear classifier and
 79 as a CAA steering vector during generation.

80 **Findings.** The direction at Layer 21 achieves 97.5% test accuracy (AUC = 0.999), but has 0.93 cosine
 81 similarity with a text length direction — it is almost entirely a verbosity detector. After projecting
 82 out length, a residual style direction still achieves 85.5% accuracy at Layer 33. Causal steering
 83 qualitatively shifts output style but with modest effect: an LLM judge found only a ~ 1 point shift on
 84 a 7-point scale, attributed to the base model’s high AI-likeness floor.

85 3 Related Work

86 **Linear representations in residual streams.** A growing body of work in mechanistic interpretability
 87 has shown that transformer language models encode high-level concepts as linear directions in their

residual streams. Marks and Tegmark [2024] provide foundational evidence, demonstrating that truth and falsehood are linearly represented and extractable via difference-in-means probing — the core technique we employ — with causal verification via surgical intervention in the forward pass. Arditi et al. [2024] extend this to behavioral properties, showing that refusal in safety-tuned models is mediated by a single direction whose ablation disables refusal and whose addition induces it on harmless inputs. Rimsy et al. [2024] introduce Contrastive Activation Addition (CAA), the methodology directly underlying our work, which extracts steering vectors by averaging residual stream differences between contrastive pairs and applies them at inference time with a tunable coefficient. Konen et al. [2024] further demonstrate that style properties — sentiment, emotion, writing style — can be extracted and steered via activation engineering without any training. Our work asks whether “sounding like AI” is similarly encoded, extending this line of research to a stylistic property with direct practical implications for detection evasion.

AI text detection and evasion. Mitchell et al. [2023] introduce DetectGPT, a zero-shot detector exploiting the observation that LLM-generated text occupies negative curvature regions of a model’s log probability function. Sadasivan et al. [2023] stress-test the robustness of a broad range of detectors — including watermarking, neural network-based, and zero-shot approaches — showing empirically and theoretically that existing methods are unreliable under paraphrasing attacks. Our approach is fundamentally different from such surface-level evasion: rather than modifying outputs after generation, we intervene directly on the model’s internal style representation during the forward pass.

Residual stream features for AI text detection. Most closely related to our work, Kuznetsov et al. [2025] apply Sparse Autoencoders (SAEs) to Gemma-2-2b’s residual stream to identify features contributing to AI text detection, finding that style and verbosity features are the most discriminative. They also find, consistent with our results, that text length is a major confounding factor. Our work differs in three key ways: we use the simpler difference-in-means approach rather than SAEs; we target evasion rather than detection interpretability; and we evaluate against real-world detectors including Pangram.

4 Methodology

Our methodology extends the preliminary study’s pipeline along three axes: replacing the LLM judge with real AI detectors, testing the instruct variant of the base model, and building a multi-source dataset. We describe each in turn.

4.1 Models

We run all experiments on **Qwen2.5-3B** (base) and **Qwen2.5-3B-Instruct**, loaded in float16 precision. For the model size experiment, we additionally run the instruct pipeline on Qwen2.5-7B-Instruct and Qwen2.5-14B-Instruct. Activations are extracted from the last-token position of the residual stream at all 37 layer outputs (embedding layer + 36 transformer layers), with a maximum input length of 512 tokens.

4.2 Direction Extraction and Classification

We follow the CAA methodology of the preliminary study exactly. For each layer ℓ , the AI direction is computed as the difference-in-means over the training split:

$$\mathbf{d}_\ell = \frac{1}{N} \sum_{i=1}^N \mathbf{a}_\ell^{(\text{AI}, i)} - \frac{1}{N} \sum_{i=1}^N \mathbf{a}_\ell^{(\text{human}, i)}$$

Classification is performed by projecting held-out activations onto the normalized direction and thresholding at the class midpoint. The best layer is selected by validation accuracy. We also compute length-orthogonal directions by projecting out a separately computed length direction (derived from a median split on character count) and sweeping across all layers for the best length-controlled accuracy.

4.3 Multi-Source Dataset

To test whether the AI direction generalizes beyond ChatGPT, we collected responses from **Claude** (claude-sonnet-4-5) and **Gemini** (gemini-2.5-flash) to the same HC3 questions via OpenRouter, yielding approximately 350 pairs per source. We removed the upper character length filter from the preliminary study, as Claude and Gemini responses are substantially more verbose than ChatGPT (mean lengths of 1,245 and 1,741 characters respectively, vs. 939 for ChatGPT). We run the direction-finding pipeline separately for each source, and also construct a pooled dataset combining all three sources (704 pairs after filtering: 329 ChatGPT, 288 Claude, 87 Gemini) to find a shared AI direction with the human side held constant.

Source	Pairs	Human mean length (chars)	AI mean length (chars)
ChatGPT	350	506	939
Claude	350	506	1,245
Gemini	347	508	1,741
Combined	704	525	1,052
Combined (truncated)	1,031	513	501

Table 1: Dataset statistics per source. Human mean lengths are nearly identical across sources since all draw from the same HC3 questions; the large variation in AI mean length reflects differences in verbosity across models. The combined dataset (704 pairs) applies stricter filtering than the individual pipelines, disproportionately reducing Gemini pairs (329 ChatGPT, 288 Claude, 87 Gemini). The truncated combined dataset retains all 1,031 pairs (345/344/342 per source) by sentence-level truncation of AI responses to match human length, nearly eliminating the length gap.

4.4 AI Detector Evaluation

Rather than using an LLM judge, we evaluate steered outputs against three commercial detectors: **Pangram** (v3 API, ai_assistance_score), **GPTZero** (completely_generated_prob), and **ZeroGPT** (fakePercentage/100). All three return a scalar probability in $[0, 1]$. GPTZero was evaluated manually via web interface due to API cost. We evaluate five steering multipliers (two negative, baseline, two positive) across thirty prompts sampled from the HC3 test set for the base model and truncated combined dataset experiments and five fixed prompts for remaining experiments, scoring each generated output independently.

4.5 Prompting Baseline

To assess whether instruction prompting achieves comparable evasion to activation steering, we prepend one of three “write like a human” instructions to each prompt and score the outputs with Pangram. We run this on both the base and instruct models: the base model serves as a sanity check (Qwen2.5-3B does not reliably follow instructions), while the instruct model provides a fairer comparison against CAA steering.

4.6 Causal Steering

Steering is implemented via a forward hook registered on the target transformer layer. At each forward pass during generation, the hook adds $\alpha \cdot \hat{\mathbf{d}}_\ell$ to the residual stream at every token position, where α is the steering multiplier and $\hat{\mathbf{d}}_\ell$ is the normalized direction scaled by its raw magnitude. Generation uses temperature 0.7, top-p 0.9, and a maximum of 150 new tokens.

5 Experiments and Results

5.1 Direction Finding: Base Model Replication

We first replicate the preliminary study’s direction-finding pipeline on the HC3 ChatGPT dataset using Qwen2.5-3B. Figure 1 shows classification performance and PCA cluster separation across all 37 layers. The direction-finding results closely match the preliminary study: the best layer by

validation accuracy is Layer 21, with test accuracy of 97.5% (AUC = 0.999, 95% CI: [0.950, 0.995]). Cross-layer direction consistency is high (mean adjacent cosine similarity: 0.874), confirming that the direction stabilizes early and remains stable throughout the network. One notable feature is a consistent dip in classification accuracy around layers 8–10 before recovering sharply at layer 11, visible in both the accuracy and silhouette plots; the cause of this is unclear.

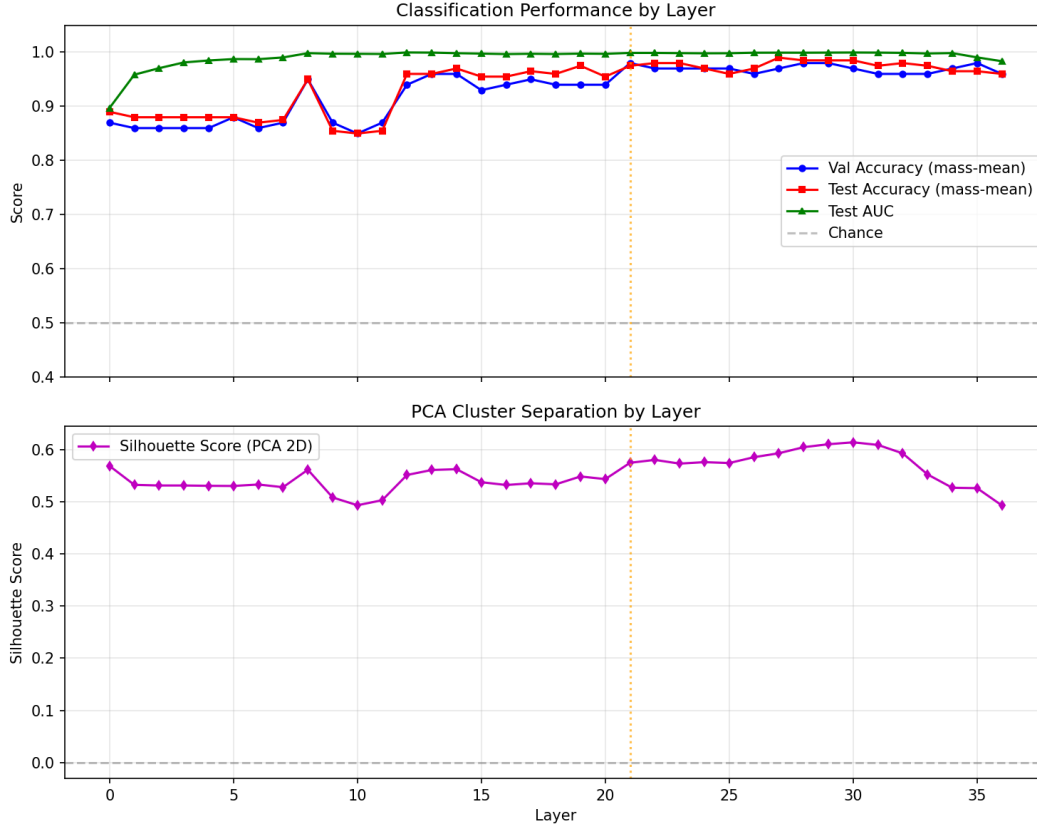


Figure 1: Classification performance (top) and PCA cluster separation (bottom) by layer for Qwen2.5-3B on the HC3 ChatGPT dataset. The orange dotted line marks the best layer (Layer 21). A consistent dip around layers 8–10 is visible in both metrics.

The length confound is substantial: cosine similarity between the AI direction and the length direction is 0.930, and accuracy drops from 97.5% to 82% after projecting out length. The best length-orthogonal direction is found at Layer 33 with 85.5% accuracy, consistent with the preliminary study.

5.2 Detector Evasion: Base Model

Replacing the LLM judge with Pangram yields a substantially stronger evasion result than the preliminary study suggested. Figure 2 shows Pangram AI probability across five steering multipliers. At the most human-steered condition (multiplier −33), Pangram assigns only 2% AI probability — effectively fooled. The baseline (multiplier 0) already scores 0.70, confirming that Qwen2.5-3B outputs are distinctly AI-sounding even without steering.

5.3 Instruct Model

Running the same pipeline on Qwen2.5-3B-Instruct yields a different best layer (Layer 24, test accuracy 97.5%, AUC = 0.998) and a different best length-orthogonal layer (Layer 19, 86% accuracy). Despite the instruct model having a more pronounced AI style — baseline Pangram score of 0.99 vs. 0.70 for the base model — steering is substantially less effective. The most human-steered condition

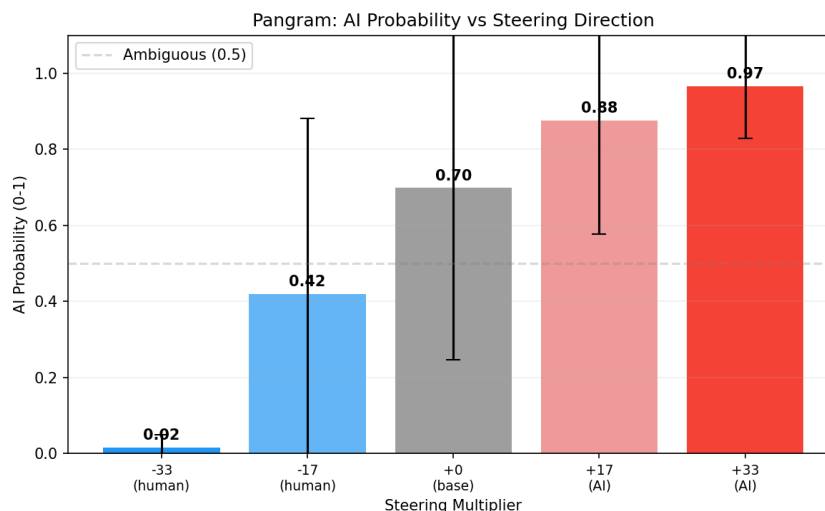


Figure 2: Pangram AI probability by steering multiplier for Qwen2.5-3B steered with the ChatGPT-derived AI direction. Negative multipliers steer toward human-like text; positive multipliers steer toward AI-like text. Error bars show standard deviation across 30 prompts.

185 only reaches 0.44, compared to 0.02 for the base model. The more entrenched AI style of the instruct
 186 model appears to make it harder to steer away from, rather than providing a larger dynamic range as
 187 initially hypothesized.

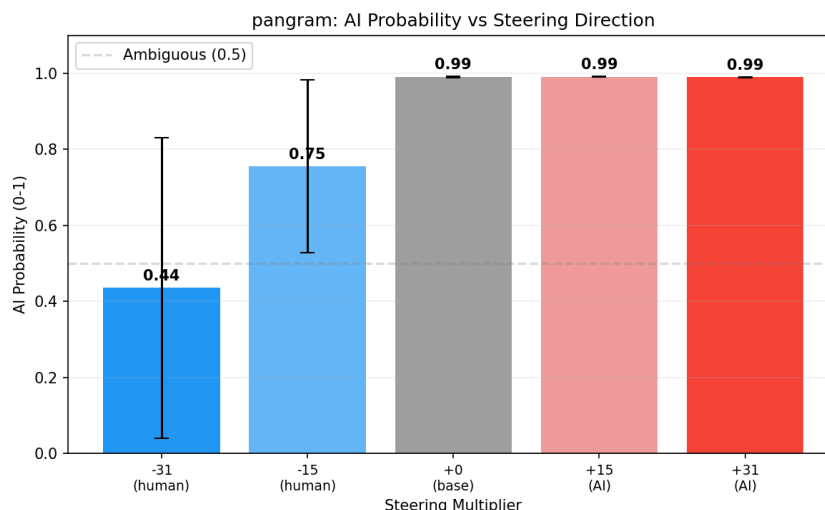


Figure 3: Pangram AI probability by steering multiplier for Qwen2.5-3B-Instruct. Compared to the base model (Figure 2), the baseline is higher (0.99 vs. 0.70) and the most human-steered condition reaches only 0.44 vs. 0.02, suggesting the instruct model’s stronger AI style is harder to override via activation steering.

188 5.4 Does the AI Direction Generalize Across Sources?

189 Table 2 shows direction-finding results across all three sources. All three achieve high classification
 190 accuracy, but the length confound is even more severe for Claude and Gemini than for ChatGPT.

191 After projecting out length, Claude and Gemini drop to 62–64% accuracy, barely above chance,
 192 compared to ChatGPT’s 80.5%. The within-class correlations tell the same story: Gemini’s correlation
 193 between AI text length and projection onto the AI direction is essentially 0 (0.004), meaning Gemini

Source	Best Layer	Test Acc	Test AUC	Cos sim w/ length	Acc (orig)	Acc (no length)
ChatGPT	22	98.5%	1.000	0.938	98.5%	80.5%
Claude	28	98.0%	0.999	0.975	98.0%	62.0%
Gemini	35	97.4%	0.996	0.961	97.4%	64.4%

Table 2: Direction-finding results per source. The length confound is severe across all sources, but especially for Claude and Gemini, whose length-orthogonal directions barely exceed chance.

194 responses are uniformly long regardless of the question. Claude is similar (0.060). This suggests that
195 for Claude and Gemini, the learned direction is almost entirely a verbosity detector with little residual
196 style signal.

197 Table 2 reports classification accuracy per source; Table 3 reports pairwise cosine similarities between
198 the extracted directions, both raw and length-orthogonal.

Pair	Cosine Sim (raw)	Cosine Sim (length-orthogonal)
ChatGPT vs. Claude	0.504	0.467
ChatGPT vs. Gemini	0.151	0.197
Claude vs. Gemini	0.430	0.251

Table 3: Pairwise cosine similarities between per-source AI directions, both raw (best layer per source) and after projecting out the length component. ChatGPT and Claude share moderate overlap; ChatGPT and Gemini are nearly orthogonal, suggesting their directions capture largely source-specific rather than universal style signals.

199 The low cross-source similarities (Table 3) are consistent with the evasion results in Figure 4:
200 ChatGPT and Claude share moderate directional overlap (0.50 raw), while ChatGPT and Gemini
201 are nearly orthogonal (0.15). Notably, the length-orthogonal similarities are lower for both Claude
202 pairs than the raw similarities, suggesting that once the shared verbosity signal is removed, what
203 remains is even more source-specific. This is reflected in the evasion results: ChatGPT’s direction
204 gives the cleanest evasion (dropping to 0.19 at most human), while Claude and Gemini directions
205 are far less effective (0.75 and 0.85 respectively). The baseline scores for Claude and Gemini are
206 both 0.99, reflecting their high verbosity. One anomaly is that Gemini’s “Most AI” condition actually
207 drops to 0.84, lower than its baseline — likely an artifact of incoherent outputs at high multipliers
208 being misclassified as human by Pangram.

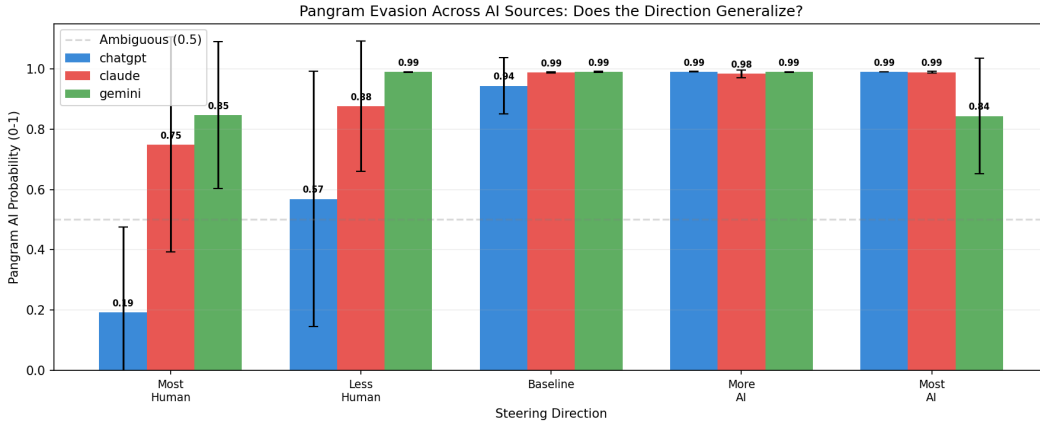


Figure 4: Pangram AI probability by steering direction and source. Each source is steered with its own best-layer direction. ChatGPT’s direction produces the cleanest evasion; Claude and Gemini directions show little effect, consistent with their directions being dominated by length.

209 5.5 Pooled Dataset

210 Combining all three sources into a pooled dataset (704 pairs) and finding a single shared direction
 211 gives a best layer of 33 with 97.1% test accuracy (AUC = 0.995). The length confound remains
 212 strong (cosine similarity 0.944), and accuracy drops from 97.1% to 70.1% after removing length —
 213 worse than ChatGPT alone but better than Claude or Gemini individually. Steering on the pooled
 214 direction produces a reversed result: negative multipliers (meant to steer human) score *higher* on
 215 Pangram than positive multipliers, the opposite of what the ChatGPT-only pipeline found. Inspecting
 216 individual outputs reveals that the effect is driven by a few outlier prompts dropping dramatically
 217 (e.g. to 0.005–0.067) while most prompts remain near 0.99 regardless of multiplier. We attribute
 218 the direction flip to the pooled “AI” centroid being pulled differently when mixing three sources
 219 with very different length distributions, potentially causing the mass-mean direction to point in the
 220 opposite orientation compared to the ChatGPT-only direction.

221 5.6 Truncation Experiment

222 To directly test whether a genuine style signal exists independent of length, we truncate AI responses
 223 sentence-by-sentence to match human response lengths (Section 4). This reduces the cosine similarity
 224 between the AI direction and the length direction from 0.944 to 0.002, and accuracy before and after
 225 removing length is identical (95.8%), confirming that the learned direction is now capturing style
 226 rather than verbosity. Direction finding on the truncated dataset gives a best layer of 29 with 95.8%
 227 test accuracy (AUC = 0.996, 95% CI: [0.935, 0.977]).

228 Figure 5 shows Pangram evasion results on the truncated dataset evaluated over 30 prompts. The
 229 direction flip observed in earlier 5-prompt evaluations is no longer present: steering now follows the
 230 expected monotonic pattern, with the most human-steered condition reaching 0.26 and the baseline
 231 at 0.78. This suggests the reversal in the 5-prompt results was a sampling artifact rather than a true
 232 property of the pooled direction. Evasion is less complete than the ChatGPT-only result (0.26 vs.
 233 0.02), consistent with the pooled direction capturing a noisier, more averaged style signal across three
 234 sources. Variance across prompts remains high, reflecting the heterogeneity of the combined dataset.

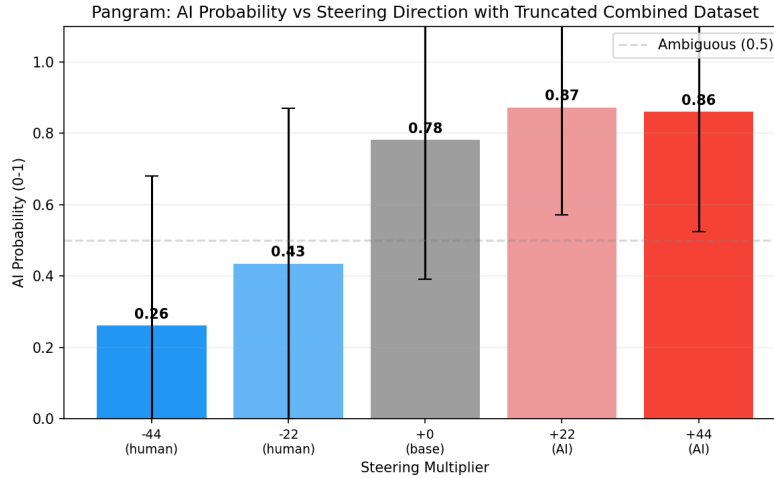


Figure 5: Pangram AI probability by steering multiplier on the truncated pooled dataset, evaluated over 30 prompts. The direction flip seen in earlier 5-prompt evaluations (see appendix) is absent; steering follows the expected monotonic pattern (0.26 → 0.43 → 0.78 → 0.87 → 0.86). Evasion is less complete than the ChatGPT-only result but genuine, with high variance reflecting the heterogeneity of the combined dataset.

235 5.7 Prompting Baseline

236 Figure 6 shows the comparison between CAA steering and prompting baseline on the instruct model.
 237 All three instruction variants flatline at approximately 0.99 Pangram score, identical to the unsteered
 238 baseline. CAA steering reduces the score to 0.44 at the most human-steered condition, substantially

outperforming prompting. On the base model, prompting similarly has no effect, as expected given that Qwen2.5-3B does not follow instructions reliably.

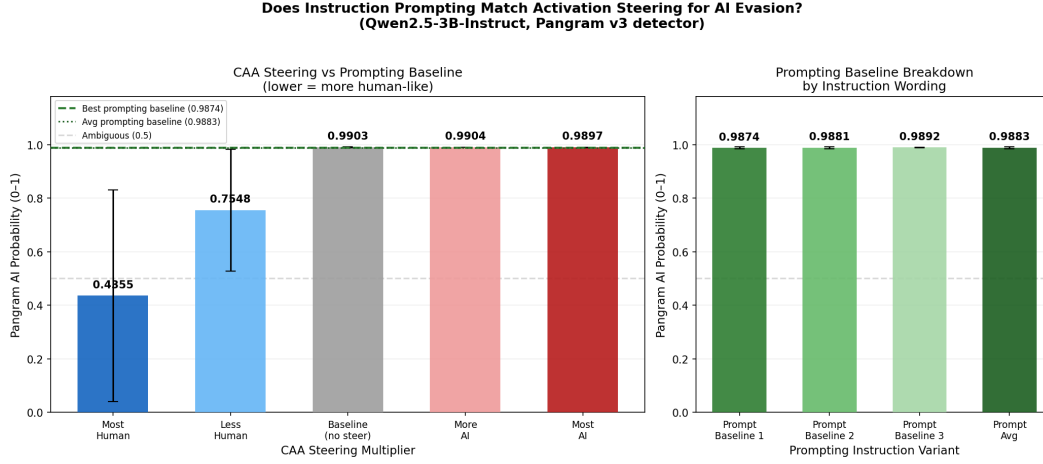


Figure 6: CAA steering vs. prompting baseline on Qwen2.5-3B-Instruct (Pangram v3). Left: steering multipliers with prompting baseline overlaid as dashed lines. Right: breakdown by instruction wording. All prompting variants flatline near 0.99; CAA steering achieves 0.44 at the most human-steered condition.

5.8 Effect of Model Size

To test whether model size affects the steerability of the instruct model, we ran the CAA pipeline on Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, and Qwen2.5-14B-Instruct using the ChatGPT-derived direction. Figure 7 shows Pangram AI probability across steering conditions for all three model sizes.

All three models have identical baselines (0.993), but steering effectiveness decreases with model size: the 3B model reaches 0.219 at the most human-steered condition, while the 7B and 14B models only reach 0.741 and 0.605 respectively. The 14B model being slightly more steerable than 7B is unexpected and may be an artifact of the small sample size (5 prompts per condition). The general trend suggests that larger instruct models are harder to steer away from AI-sounding outputs via activation addition, consistent with the hypothesis that larger models have more deeply entrenched stylistic patterns from RLHF fine-tuning.

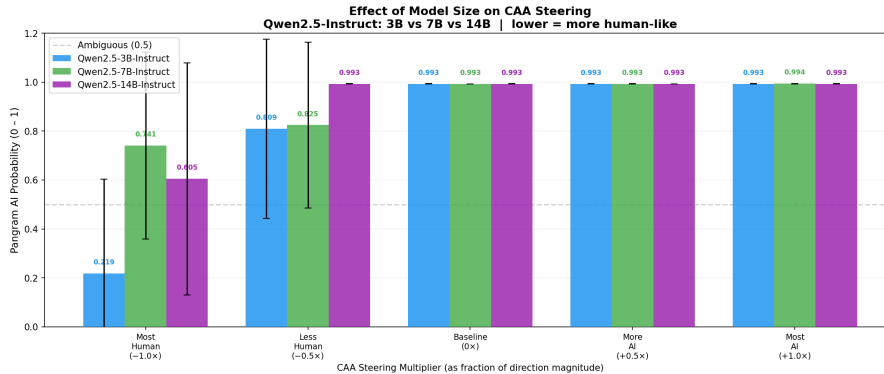


Figure 7: Effect of model size on CAA steering effectiveness (Pangram v3, 5 prompts per condition). All three Qwen2.5-Instruct models share an identical baseline of 0.993, but the 3B model is substantially more steerable than the 7B and 14B models. Note that results are based on only 5 prompts per condition and should be interpreted with caution.

6 Discussion & Conclusion

The length confound dominates, but a real style signal exists. The most consistent finding across all experiments is that the AI direction is overwhelmingly a verbosity detector. This is most extreme for Claude and Gemini, where over 96% of the learned direction aligns with text length, and length-orthogonal accuracy barely exceeds chance. The truncation experiment provides the clearest evidence that a genuine style signal exists beyond length: after eliminating the length gap, classification accuracy remains at 95.8%, indicating the model encodes real stylistic differences between AI and human text. However, this style signal is harder to exploit for evasion than the length-confounded direction.

A pooled cross-source direction produces real but partial evasion. The per-source results show that ChatGPT’s direction produces the cleanest evasion (0.02), while Claude and Gemini’s own directions are largely ineffective (0.75 and 0.85 respectively), consistent with their directions being dominated by length. The pooled direction, evaluated on the length-controlled dataset with 30 prompts, achieves intermediate evasion (0.26) — better than either Claude or Gemini individually, but not as strong as ChatGPT alone. This suggests that pooling partially recovers a shared style signal that is obscured in the Claude and Gemini per-source directions by verbosity, but that ChatGPT’s style signal remains the dominant contributor to cross-source evasion.

Unsurprisingly, activation steering outperforms prompting. The prompting baseline averages around 0.99 on both the base and instruct models with very little variation, while CAA steering achieves 0.02 on the base model and 0.44 on the instruct model. The base model result is somewhat unsurprising since Qwen2.5-3B does not follow instructions reliably, but the instruct model result is more telling: even with a model that can follow instructions, explicit prompting to “write like a human” has no measurable effect on Pangram scores. This suggests that whatever Pangram is detecting is not easily overridden by surface-level stylistic intent, but can be partially overridden by intervening directly on internal representations.

Limitations. Several limitations affect the strength of our conclusions. First, the majority of steering evaluations use only 5 prompts per condition, which is too few to draw robust conclusions — a single incoherent or anomalous output can substantially shift the mean, as seen in the pooled and truncated dataset results. While initial evaluations used 5 prompts per condition due to API cost constraints, key experiments were subsequently rerun with 30 prompts; remaining experiments retain the 5-prompt evaluation. Second, GPTZero and ZeroGPT results were noisy and non-monotonic, possibly due to the short length of steered outputs (both detectors noted unreliability on inputs less than 250 words); we were unable to draw reliable conclusions from either detector. Finally, all experiments use a single model family (Qwen2.5); generalization to other architectures remains untested.

Conclusion. We extended a preliminary study on mechanistic interpretability of AI-generated text in three directions: multi-source dataset construction, real detector evaluation, and length control. Our results confirm that a linear AI direction exists in Qwen2.5-3B’s residual stream, and that steering with the ChatGPT-derived direction can substantially reduce Pangram detection scores. However, this direction does not generalize cleanly across AI sources, and a pooled multi-source direction fails to produce consistent evasion. The dominant confound throughout is text length, particularly for Claude and Gemini. Taken together, our findings suggest that “sounding like AI” is not a single universal direction in representation space, but a source-specific composite of style and verbosity that varies across model families.

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329 A Example Steered Outputs

330 Below are some examples of steering results, run on Qwen2.5-3B. For each prompt, we show five
331 steering conditions ranging from most human-like to most AI-like with their multipliers. Outputs are
332 truncated at the generation limit.

333 Prompt 1: Write a short paragraph about climate change:

334 **Most Human** (−33.2) The average temperature on Earth is rising. The earth’s atmosphere is getting
335 warmer. This is caused by the greenhouse effect. This greenhouse effect is the result of CO2
336 and other green house gases. This is causing the temperature to rise and rise. If we do not
337 reduce our green house gas emissions, the temperature will rise to 3 degrees C. The sea level
338 will rise 120 m. We will not be able to live on the planet. If we don’t reduce our green house
339 gas emissions, there will be a flood of 200 cm in London. The ice in the north and south
340 pole will melt and there will be a rise in sea level. The sea will flood our coasts. Our coasts
341 will [...]

342 **Less Human** (−16.6) Climate change is the long-term alteration of the statistical distribution of
343 weather patterns over several decades or longer. Climate change is a change in the statistical
344 distribution of weather over periods of time that exceed a few decades which is typically
345 associated with a change in the mean or the variability of those processes. It may be in
346 response to natural or human-induced changes in radiative forcing of the climate system, or
347 to changes in physical or biological systems. The changes in climate may be identified in
348 records of actual surface weather phenomena, in measurements of physical properties of the
349 atmosphere, or in the amount and distribution of snow and ice, or in sea level.

350 **Baseline** (0) Climate change is a global phenomenon that refers to the long-term alteration of Earth’s
351 climate patterns and weather conditions. It is primarily caused by human activities, such as
352 the burning of fossil fuels, deforestation, and industrial processes, which release greenhouse
353 gases into the atmosphere. These gases trap heat from the sun and contribute to the warming
354 of the planet. The consequences of climate change are far-reaching, including rising sea
355 levels, more frequent and intense natural disasters, and significant impacts on ecosystems
356 and biodiversity. Urgent action is needed to mitigate its effects and adapt to the changes
357 already occurring.

358 **More AI** (+16.6) Climate change refers to the long-term changes in weather patterns and temper-
359 ature that are caused by various factors, including human activities such as burning fossil
360 fuels and deforestation. These changes can have significant impacts on ecosystems, wildlife,
361 and human societies, such as rising sea levels, more frequent extreme weather events, and
362 disruptions to agriculture and water resources. Efforts to mitigate climate change involve
363 reducing greenhouse gas emissions, promoting renewable energy sources, and implementing
364 sustainable practices in various sectors of society.

365 **Most AI** (+33.2) Climate change is a pressing issue that has become increasingly important in
366 recent years. As the world continues to face the impacts of climate change, it is important
367 for individuals, governments, and organizations to take steps to reduce greenhouse gas
368 emissions and promote sustainability. This includes measures such as reducing reliance
369 on fossil fuels, investing in renewable energy sources, and promoting energy efficiency.
370 Additionally, individuals can make lifestyle changes such as reducing their carbon footprint
371 by conserving energy, using public transportation, and supporting environmentally-friendly
372 products.

373 Prompt 2: Tell me about the history of pizza:

374 **Most Human** (−33.2) Pizza is one of the most popular dishes in the world. Pizza is a round flatbread
375 baked in a pizza oven and topped with a variety of things. The flatbread is called a pizza
376 base. The pizza base is flat and round, and has a very thin base. The base is made of flour,
377 salt, yeast and water. The yeast is what makes the dough rise. It is like a balloon. The dough
378 is stretched into a circle. The dough is rolled up into a circle and baked. The baking makes
379 the dough very thin. The pizza base is then topped with cheese, ham, pepperoni, mushrooms,

380 olives, onions, whatever you like. The pizza is then cooked for a few minutes. When it is
381 cooked it [...]

382 **Less Human** (−16.6) Pizza originated in Naples, Italy, and was brought to the United States by
383 Italian immigrants. The first pizza restaurant in the US was opened in 1905 in New York.
384 The first pizza restaurant in London opened in 1919. In 1929, the first pizza parlor opened in
385 Canada. In 1936, pizza became an official meal in the US Army. In 1958, Domino's Pizza
386 was founded by Tom and Mary Simonsen. In 1960, the first pizza chain in the US opened.
387 In 1962, Pizza Hut was founded. In 1964, Papa John's opened. In 1975 [...]

388 **Baseline** (0) Pizza is a dish of Italian origin consisting of a usually round, flattened base of leavened
389 wheat-based dough topped with tomatoes, cheese, and various other ingredients, which is
390 usually baked in an oven. Pizza originated in Naples, Italy.

391 **More AI** (+16.6) Pizza is a popular dish that originated in Italy. It is believed to have been created
392 in Naples, Italy, during the 18th century. The first pizzas were made with simple ingredients
393 like tomato sauce, mozzarella cheese, and olive oil. Over time, different regions in Italy
394 developed their own unique styles of pizza, such as the Neapolitan-style pizza and the
395 Sicilian-style pizza. Pizza became popular throughout the world, and today it is enjoyed by
396 people of all ages and backgrounds.

397 **Most AI** (+33.2) Pizza is a popular and beloved dish that has a rich history dating back to ancient
398 times. The origins of pizza can be traced back to various regions around the world, but the
399 modern version of pizza as we know it today is typically associated with Italy. Here's a
400 brief overview of the history of pizza: 1. Ancient Origins: The roots of pizza can be traced
401 back to various cultures around the world, with some evidence suggesting that early forms
402 of flatbread, such as unleavened bread or flatbreads made from grains like wheat or barley,
403 were used in ancient times. 2. Early Italian Pizza: The term "pizza" comes from the Italian
404 word "pizza," which means "flatbread." The earliest versions of pizza in Italy were often [...]

405 **B Prompts**

406 Below, find the three prompts used in the prompting baseline:

- 407 1. "Write in a natural, conversational human style. Avoid sounding like an AI."
- 408 2. "Respond as a real person would: casually, directly, and without over-explaining."
- 409 3. "Write like a human, not like an AI assistant. Be informal and natural."

410 **C Additional Figures**

411 Figures 8–11 show GPTZero and ZeroGPT results for both the base and instruct models, evaluated
412 on ChatGPT-sourced steered outputs (5 prompts per condition). GPTZero was evaluated manually
413 via web interface due to API cost constraints and so was not used for subsequent experiments.

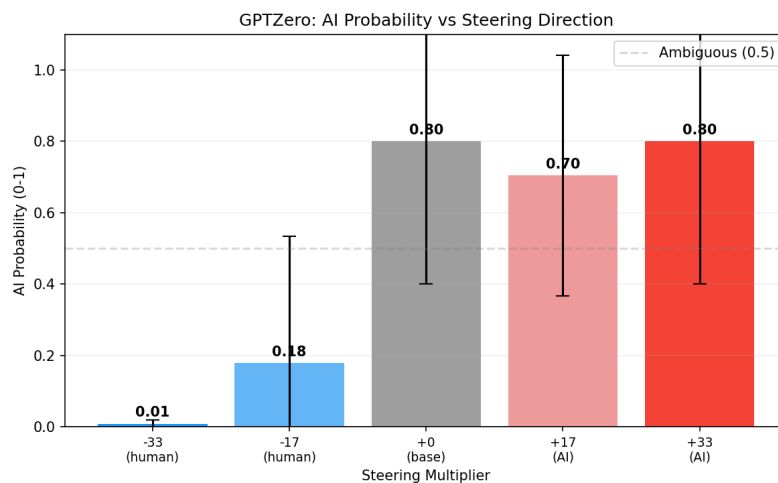


Figure 8: GPTZero AI probability by steering multiplier for Qwen2.5-3B (base). High variance across prompts and the non-monotonic positive multipliers suggest instability at 5 prompts.

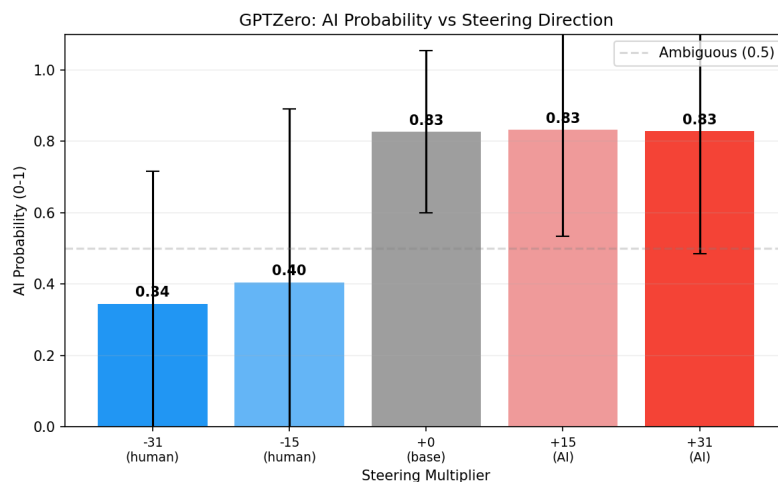


Figure 9: GPTZero AI probability by steering multiplier for Qwen2.5-3B-Instruct. The most human-steered condition reaches 0.34, but scores are flat from baseline onward (0.83 across all positive multipliers), suggesting GPTZero is largely insensitive to AI-ward steering on the instruct model.

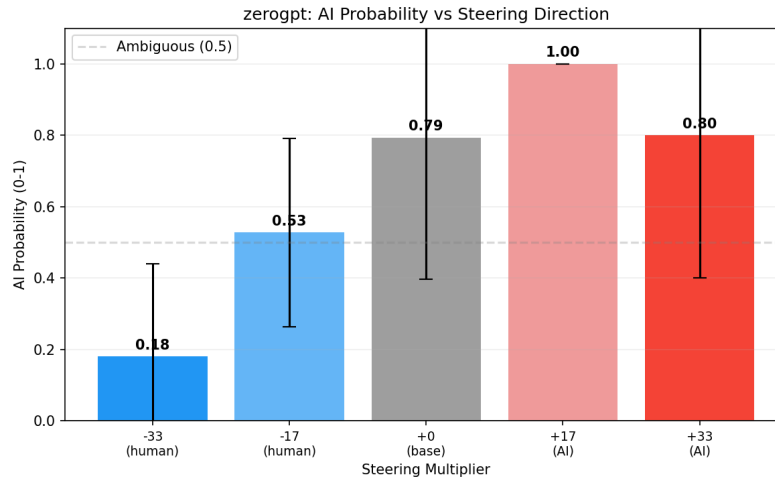


Figure 10: ZeroGPT AI probability by steering multiplier for Qwen2.5-3B (base). Results are non-monotonic: the most human-steered condition reaches 0.18, but the less human-steered condition spikes to 1.00, and the most AI-steered condition drops to 0.80. ZeroGPT appears unreliable on these short outputs (under 150 words), consistent with the detector’s own guidance on minimum input length.

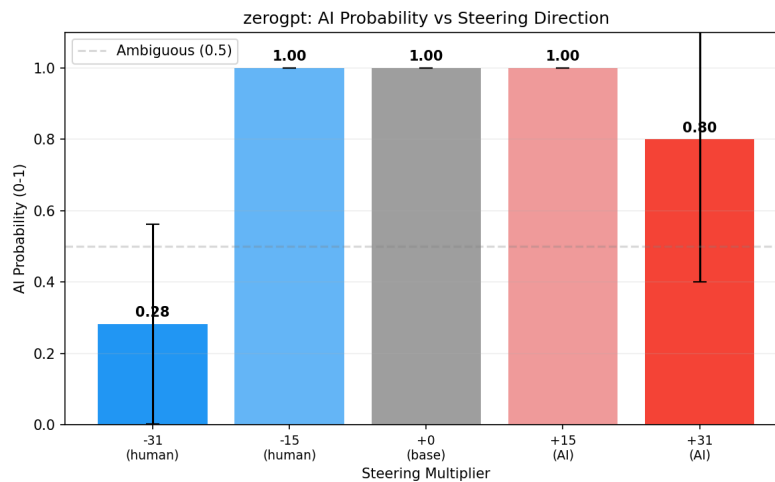


Figure 11: ZeroGPT AI probability by steering multiplier for Qwen2.5-3B-Instruct. The most human-steered condition reaches 0.28, but the pattern is again non-monotonic, with the less human condition jumping to 1.00 and the most AI condition dropping to 0.80. As with the base model, results are unreliable at this output length.

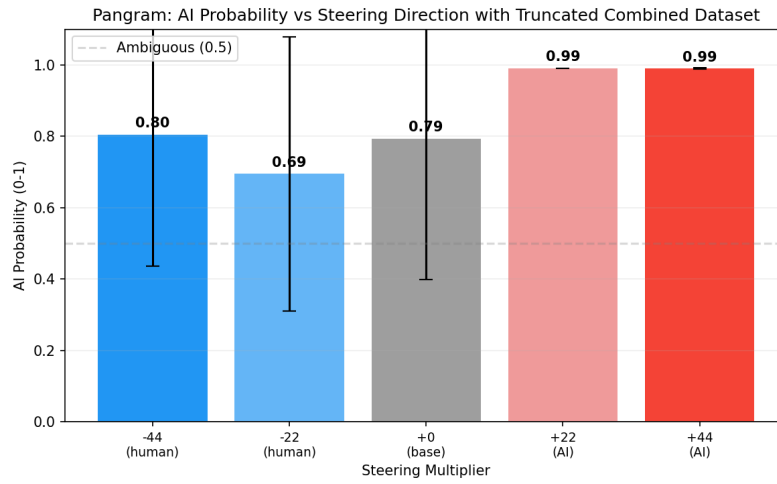


Figure 12: Pangram AI probability by steering multiplier on the truncated pooled dataset, evaluated over 5 prompts. The direction appears reversed relative to the ChatGPT-only result, with negative (human-steered) multipliers scoring higher than positive (AI-steered) multipliers. As shown in Section 5.6, this reversal disappears when evaluated over 30 prompts, indicating it is a sampling artifact rather than a true property of the pooled direction.